Q31 ALY 3510 · Harder · Question

A swarm of individually simple agents reroutes around failures and still produces a coordinated solution, even though no agent has a map of the whole system. Which conclusion is best supported?

A. The system must contain a hidden central controller

B. Collective intelligence can emerge from local interactions without centralized awareness

C. Each agent possesses a complete model of the problem

D. The result proves that the group is consciously aware

Q32 ALY 3510 · Harder · Question

A hiring recommendation is produced by several connected models operated by different vendors. A rejected applicant challenges the outcome. Which concerns are most directly raised? (Select ALL that apply)

A. Responsibility may be difficult to assign across the participating organizations

B. Data permissions and privacy obligations may differ across the connected systems

C. Distribution guarantees that no biased output can occur

D. Auditing the full decision path may require cooperation from several parties

Q33 ALY 3510 · Harder · Question

A hospital keeps patient data locally but shares model updates with partner hospitals. Which interpretation is most accurate?

A. No information can ever be inferred from the shared updates

B. The approach may reduce direct sharing of raw data while still creating privacy and governance concerns

C. The system is centralized because all hospitals use the same goal

D. Local data storage eliminates the need for security controls

Q34 ALY 3510 · Harder · Question

A domain expert uses an AI framework to prototype a failure-prediction model without implementing backpropagation or GPU scheduling. What has changed, and what has not? (Select ALL that apply)

A. The framework lowers the barrier between a domain idea and a working prototype

B. The domain expert can focus more on defining failure and selecting meaningful data

C. Technical review becomes unnecessary because the framework handles the implementation

D. Interdisciplinary collaboration becomes easier because reusable components provide a shared starting point

Q35 ALY 3510 · Harder · Question

Two teams use the same AI framework and training library but obtain very different model quality. Which explanation best fits the role of frameworks?

A. Frameworks standardize tools, not the quality of data, problem definition, or evaluation

B. Using the same framework guarantees equivalent results

C. The lower-performing team must have used symbolic AI

D. Frameworks eliminate the effect of domain expertise

Q36 ALY 3510 · Harder · Question

An NLP system translates a technically correct phrase but removes a culturally meaningful implication. What does this result show?

A. Grammar alone is sufficient to preserve meaning

B. Effective translation requires more than literal word substitution

C. The source language cannot be processed computationally

D. A larger vocabulary always solves cultural ambiguity

Q32 ALY 3510 · Harder · Answer

Answers: A, B, D

Concept: Distributed governance

Three companies' models made the call together. Who's responsible? Whose privacy rules apply? Tracing the decision means getting all of them to cooperate. C is wrong: splitting up the work doesn't remove bias. "Guarantees" is the giveaway.

Q31 ALY 3510 · Harder · Answer

Answer: B

Concept: Distributed AI and emergence

Think of ants finding food. No ant has the map, but by following simple rules with their neighbors, the colony finds the path. Smart group behavior doesn't need a boss. It also doesn't prove the group is conscious.

Q34 ALY 3510 · Harder · Answer

Answers: A, B, D

Concept: AI frameworks and domain expertise

The framework is like a power tool. You can build something without knowing how the motor is wired, and spend your time on what counts as a failure. Someone still has to check the work, so C is wrong. "Unnecessary" is the giveaway.

Q33 ALY 3510 · Harder · Answer

Answer: B

Concept: Federated learning

Each hospital keeps its records at home and shares only what its model learned. Less raw data moves around, but the shared updates can still leak hints, so privacy and security still matter. A and D say "never" and "eliminates."

Q36 ALY 3510 · Harder · Answer

Answer: B

Concept: NLP and culture

The words were right, but the meaning got lost, like translating a joke word for word. Good translation needs culture and context, not just swapping one word for another.

Q35 ALY 3510 · Harder · Answer

Answer: A

Concept: AI frameworks

Two cooks, same oven, very different dinners. The oven is the framework. The ingredients (data), the recipe (problem definition), and the taste test (evaluation) still decide how it turns out.

Q37 ALY 3510 · Harder · Question

A language model generates a convincing apology but has no personal experience, intention, or emotional stake in the situation. Which question does this most directly raise?

A. Whether useful communication requires genuine intention and understanding

B. Whether machines can process text quickly

C. Whether all communication must be grammatically formal

D. Whether vocabulary size determines consciousness

Q38 ALY 3510 · Harder · Question

Which claims are supported by the success of large language models? (Select ALL that apply)

A. Sophisticated language behavior can emerge from learned statistical patterns

B. Fluent output alone proves human-like understanding

C. Explicitly programming every fact and grammar rule is not required for useful language generation

D. The boundary between convincing language use and genuine understanding remains unsettled

Q39 ALY 3510 · Harder · Question

A sentiment system marks "Perfect, another two-hour outage" as positive. What combination of limitations most likely explains the error?

A. It relied too heavily on a positive word and lacked enough situational context to recognize sarcasm

B. It used too much data and therefore became deductive

C. It correctly recognized the reviewer's literal intention

D. It performed model parallelism instead of data parallelism

Q40 ALY 3510 · Harder · Question

A multilingual chatbot performs well in English but poorly in a low-resource language. Which responses are reasonable? (Select ALL that apply)

A. Examine whether training data quantity and quality differ across the languages

B. Evaluate results with speakers who understand the language and cultural context

C. Assume the same thresholds and error patterns transfer unchanged across languages

D. Treat the performance gap as a potential access and fairness issue

Q41 ALY 3510 · Harder · Question

A company uses an LLM to draft public safety instructions. The text is fluent and confident but contains a factual error. What is the most appropriate conclusion?

A. Fluency is evidence of reliable factual understanding

B. The output should be accepted because pattern recognition is usually accurate

C. Coherent language and factual reliability are separate qualities that require verification

D. The error proves that statistical models cannot produce useful text

Q42 ALY 3510 · Harder · Question

An AI-generated audio message accurately imitates a trusted executive and requests an urgent transfer. Which risks are present? (Select ALL that apply)

A. Deception about whether the communication came from a human source

B. Misinformation or fraud delivered through realistic synthetic media

C. Guaranteed identification by automated AI detectors

D. Questions about consent and acceptable use of a person's voice

Q38 ALY 3510 · Harder · Answer

Answers: A, C, D

Concept: LLMs and understanding

LLMs show that good writing can come from learned patterns, with nobody hand-coding every rule. Whether that counts as understanding is still open. B is wrong: "proves" goes too far.

Q37 ALY 3510 · Harder · Answer

Answer: A

Concept: Language and intention

The apology sounds sincere, but nobody is actually sorry. So: does good communication need real meaning and intent behind it, or are the right words enough?

Q40 ALY 3510 · Harder · Answer

Answers: A, B, D

Concept: Global NLP and fairness

Less training data in a language usually means worse results. Check with native speakers, and treat the gap as a fairness problem. C is wrong: what works in English can't be assumed to work everywhere.

Q39 ALY 3510 · Harder · Answer

Answer: A

Concept: Sarcasm and sentiment

It saw "Perfect" and stopped there. A person knows nobody is happy about a two-hour outage. Catching sarcasm takes context that the word alone doesn't give you.

Q42 ALY 3510 · Harder · Answer

Answers: A, B, D

Concept: Synthetic media ethics

A fake voice of your boss asking for money. It fools you about who's talking, it's fraud, and it uses someone's voice without permission. C is wrong: detectors can't always catch it. "Guaranteed" is the giveaway.

Q41 ALY 3510 · Harder · Answer

Answer: C

Concept: Fluency vs. reliability

Sounding confident isn't the same as being right. A smooth-talking model can still state a wrong fact, so check anything that matters, especially safety instructions.

Q43 ALY 3510 · Harder · Question

A predictive model used for preventive maintenance misses most true failures but rarely raises a false alarm. Which evaluation concern is most important?

A. High precision may be hiding poor recall

B. High recall may be hiding poor precision

C. The model is necessarily overfitting

D. The task should be treated as regression because equipment is involved

Q44 ALY 3510 · Harder · Question

A model performs extremely well during development because information recorded after each equipment failure was accidentally included among the training inputs. What is the main problem?

A. Underfitting

B. Data leakage

C. Reinforcement learning

D. Analogical reasoning

Q45 ALY 3510 · Harder · Question

A neural network and a symbolic system both reach the same recommendation. Which statement best distinguishes how each may have reached it?

A. The neural network applies only written rules, while the symbolic system adjusts weights from data

B. The neural network may rely on learned distributed representations, while the symbolic system follows explicit facts and rules

C. Both methods must represent knowledge in exactly the same form

D. The symbolic system cannot provide any explanation because rules are hidden

Q46 ALY 3510 · Harder · Question

A model performs well on examples similar to its training set but fails when deployed at a building with different equipment and operating patterns. Which interpretation is strongest?

A. The model may not have generalized beyond the original data distribution

B. The model has become conscious of its original environment

C. The new building proves that supervised learning cannot work

D. The failure guarantees that the model was trained without labels

Q44 ALY 3510 · Harder · Answer

Answer: B

Concept: Data leakage

The model peeked at the answer key. It got information from after each failure, which it would never have in real life, so its test scores make it look much better than it really is.

Q43 ALY 3510 · Harder · Answer

Answer: A

Concept: Precision and recall

The model almost never cries wolf, so precision is high. But it misses most of the real wolves, so recall is low. In maintenance, a missed failure is usually the expensive mistake.

Q46 ALY 3510 · Harder · Answer

Answer: A

Concept: Generalization and distribution shift

It learned one building's equipment and got confused at a different one. That's distribution shift: doing well on familiar data doesn't guarantee doing well on new data.

Q45 ALY 3510 · Harder · Answer

Answer: B

Concept: Neural vs. symbolic AI

Same answer, different route. The neural network learned patterns spread across its weights. The symbolic system followed rules someone wrote down. A flips the two, which makes it the trap.